Temporal Patterns in New York City Taxi Trips: Data Cleaning, Feature Generation and Exploration

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Abstract

This report explores temporal patterns in the New York City Taxi Trip Duration training sample. The data contain 1,458,644 trips recorded between January and June 2016. Implausible durations, distances, speeds, coordinates and passenger counts were removed before analysis. The cleaning process retained 1,443,535 trips, or 98.96% of the original sample. Friday had the highest average daily trip count. Weekday demand was concentrated around the evening commute, while weekend activity shifted towards late-night hours. Memorial Day produced the largest holiday reduction, whereas Valentine's Day was busier than a typical Sunday. Average trip speed was highest at 05:00 and lowest during the congested daytime period.

1 Introduction

The New York City Taxi Trip Duration dataset contains pickup and drop-off times, coordinates, passenger counts and trip durations for trips recorded between 1 January and 30 June 2016 [1]. This report addresses three tasks. First, it identifies and removes implausible observations. Second, it generates distance, day-of-week and average-speed features. Third, it uses the cleaned data to examine daily, hourly, weekday, weekend and holiday patterns.

All temporal results use pickup time. Trip counts refer to the Kaggle training sample rather than every taxi trip completed in New York City. They are therefore treated as a consistent measure of activity within the sample.

2 Data cleaning

The dataset contained 1,458,644 rows and no missing values. However, several values were not plausible. The maximum recorded duration was approximately 40.8 days, the maximum straight-line distance was 1,240.9 km, and the maximum implied average speed was 9,274.8 km/h. Some coordinates were also far outside New York City, while 65 trips had passenger counts outside the valid range of one to six.

Distance was calculated using the Haversine formula:

```
d = 2R arcsin(sqrt(sin^2(dLat/2) + cos(lat1) cos(lat2) sin^2(dLon/2)))
```

where R = 6371.0088 km. This gives straight-line distance rather than road distance, but it is sufficient for detecting major coordinate and speed errors.

Table 1 summarises the cleaning rules. The limits were kept broad so that unusual but credible trips remained in the sample. A duration below 30 seconds is unlikely to represent a completed passenger trip, while a duration above four hours is more likely to indicate a timestamp or meter error. Distances below 50 metres make speed unreliable, and distances above 100 km are extreme for this setting. An average speed above 120 km/h is not credible for a taxi travelling through New York. The coordinate boundary includes Manhattan, JFK, LaGuardia and Newark while excluding distant points.

Table 1: Cleaning rules and the number of trips flagged

Criterion	Trips flagged
Duration outside 30 seconds to 4 hours	6,859
Distance outside 0.05 to 100 km	11,348
Average speed above 120 km/h	136
Pickup or drop-off outside the NYC metro box	998
Passenger count outside 1 to 6	65

The rules overlap, so their individual counts cannot be added. A trip was removed if it failed at least one rule. In total, 15,109 trips were removed, leaving 1,443,535 trips. This is a reduction of only 1.04%, so the later patterns are not produced by aggressive trimming. Figure 1 shows the raw distributions and the selected boundaries.

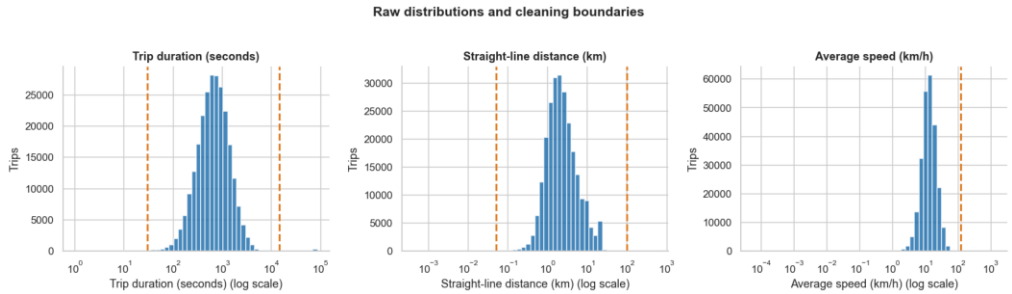


Figure 1: Raw duration, straight-line distance and average-speed distributions. The horizontal axes are logarithmic and dashed lines show the cleaning boundaries.

3 Feature generation

Three required features were generated. Trip distance was calculated in kilometres using the Haversine formula. Day of week was extracted from pickup time and stored as an ordered category from Monday to Sunday. Average speed was calculated as:

```
average speed (km/h) = distance (km) / (trip duration (seconds) / 3600)
```

Pickup date, hour, weekday number and a weekday/weekend indicator were also created. These additional fields allowed each temporal comparison to be performed directly and kept the weekday order consistent across tables and figures.

4 Time-based analysis

4.1 Most popular day of the week

Friday was the most popular day, with an average of 8,507.5 trips per day. Saturday was close behind with 8,406.9 trips. The six-month period contains 26 occurrences of every weekday, so total counts and average daily counts produce the same ranking. The average is reported because it is easier to interpret and remains appropriate for periods with unequal weekday frequencies.

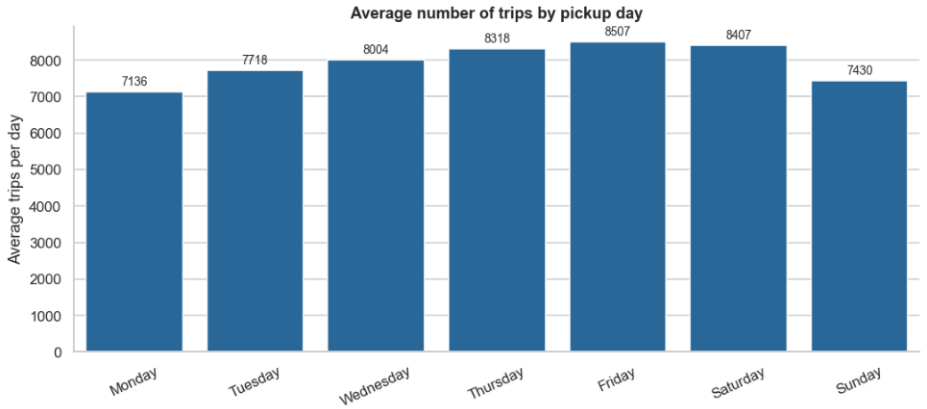


Figure 2: Average number of trips by pickup day. Friday recorded the highest daily average, followed by Saturday.

4.2 Most popular hour on each day

Figure 3 shows the share of each weekday's trips occurring in each hour. Monday and Tuesday peaked at 18:00, Wednesday at 19:00, Thursday at 21:00, Friday at 19:00, Saturday at 23:00 and Sunday at 00:00.

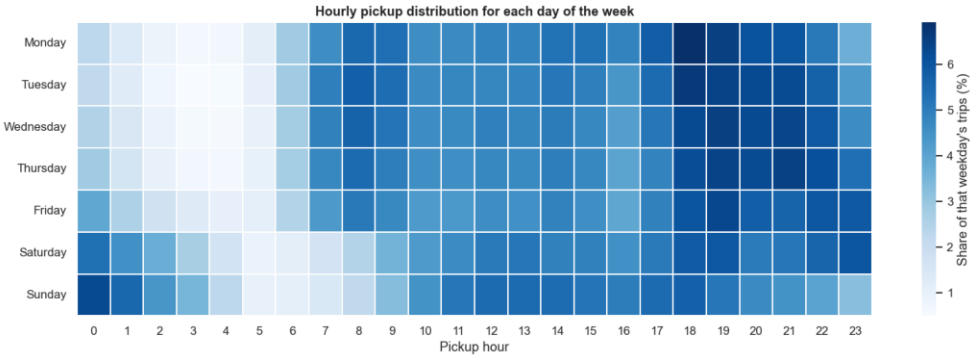


Figure 3: Hourly pickup distribution for each day of the week. Colour shows the percentage of that weekday's trips occurring in each hour.

The Monday-to-Wednesday peaks are consistent with evening commuting and after-work activity. Thursday and Friday remain busy later, while the Saturday and Sunday midnight peaks are consistent with nightlife trips crossing into the early morning. These are plausible explanations rather than confirmed trip purposes because the dataset does not record why each passenger travelled.

4.3 Weekdays and weekends

Figure 4 compares average hourly counts per day. Weekdays show clear morning and evening commuter patterns. Weekend activity begins later in the morning and is stronger between 22:00 and 01:00. Despite these different shapes, the average daily volumes are almost identical: 7,936.7 trips on a weekday and 7,918.5 on a weekend day.

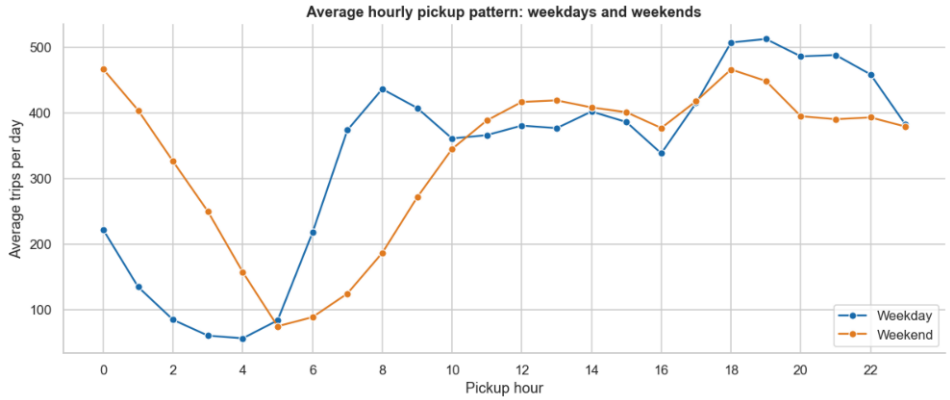


Figure 4: Average hourly pickup patterns on weekdays and weekends. Daily averaging prevents the five weekdays from outweighing the two weekend days.

Weekend trips were slightly longer in distance and faster on average. The median weekend distance was 2.22 km compared with 2.07 km on weekdays, while the mean speed was 15.84 km/h compared with 13.92 km/h. Lighter traffic and a different mix of destinations may explain these differences. However, the median weekend duration was slightly shorter, at 10.37 minutes compared with 11.38 minutes on weekdays.

4.4 Major holidays

The analysis used the correct 2016 dates: Martin Luther King Jr. Day on 18 January, Valentine's Day on 14 February, St. Patrick's Day on 17 March, Easter Sunday on 27 March and Memorial Day on 30 May. Each holiday was compared with the median hourly profile of the same weekday, excluding the five named holidays. Matching by weekday separates the holiday effect from normal weekly variation.

Table 2: Holiday trips compared with a typical matching weekday

Holiday	Trips	Typical	Change	Peak
Martin Luther King Jr. Day	7,060	7,166.5	-1.5%	18:00
Valentine's Day	8,393	7,556.0	+11.1%	17:00
St. Patrick's Day	8,167	8,324.0	-1.9%	20:00
Easter Sunday	7,290	7,556.0	-3.5%	00:00
Memorial Day	5,513	7,166.5	-23.1%	16:00

Memorial Day produced the largest change. It had 23.1% fewer trips than a typical Monday, and its peak moved from the usual 18:00 to 16:00. This is consistent with the removal of the normal weekday commute. Valentine's Day had 11.1% more trips than a typical Sunday and peaked at 17:00 rather than midnight, which may reflect additional early-evening social travel. Easter was 3.5% quieter than a typical Sunday. Martin Luther King Jr. Day and St. Patrick's Day had daily totals close to their comparison days, although their hourly shapes still changed.

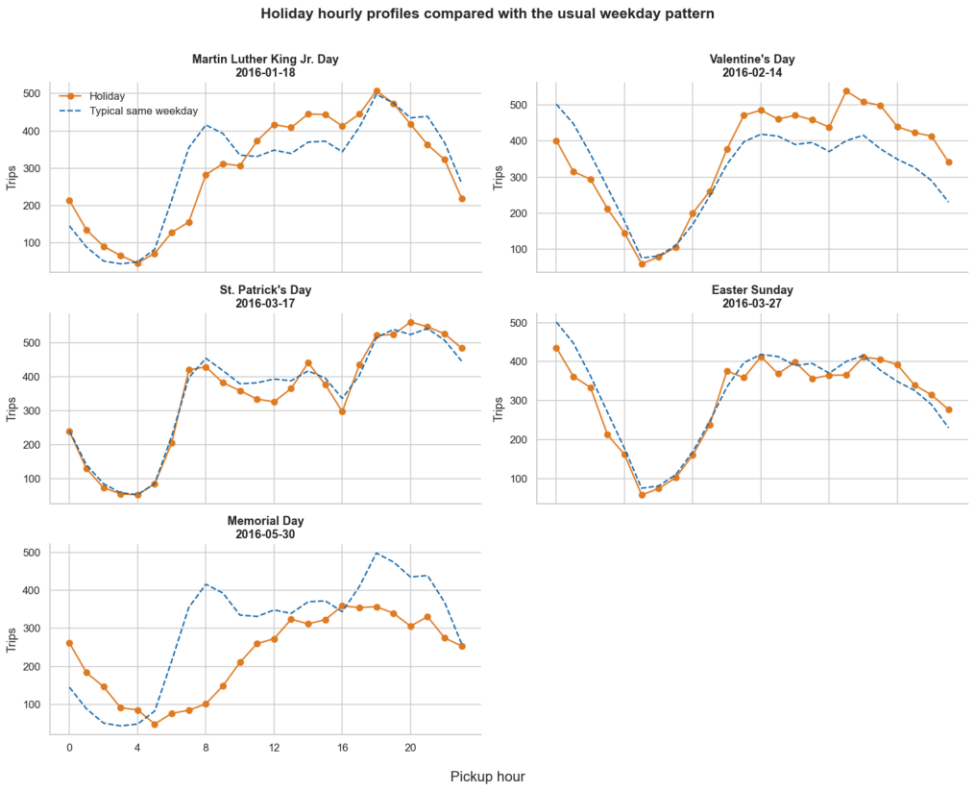


Figure 5: Hourly holiday profiles compared with the median profile of the same weekday. Holiday effects differ in both total volume and timing.

4.5 Average speed throughout the day

Figure 6 shows the mean and median trip speed by pickup hour. Trips were fastest at 05:00, when the mean speed was 24.58 km/h and the median was 22.84 km/h. Speed fell sharply from 06:00 as morning activity increased and remained low through the working day. The lowest mean occurred at 15:00, at 12.14 km/h. Speed then increased after 19:00.

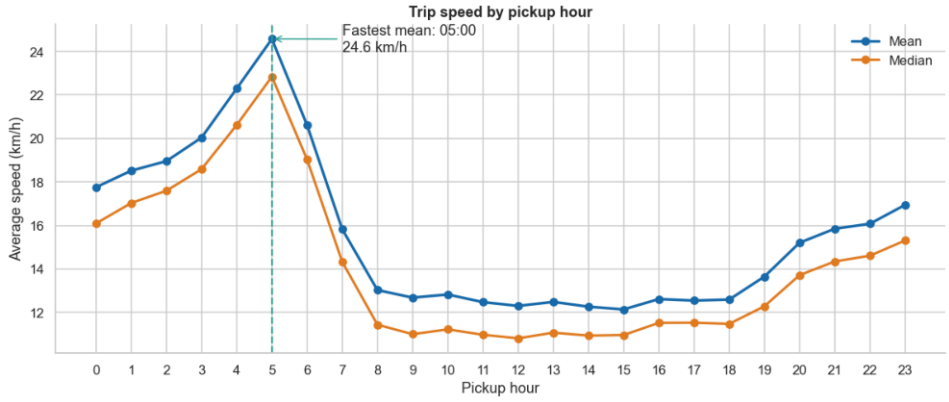


Figure 6: Mean and median average speed by pickup hour. Both measures show the same daily pattern.

The agreement between the mean and median suggests that the pattern is not caused only by a small number of unusually fast trips. Traffic congestion is the most likely explanation, although route mix and trip distance may also change during the day.

5 Limitations

The Haversine distance measures straight-line displacement and therefore underestimates the road distance travelled. The training data are a sample of recorded trips rather than a complete measure of taxi demand. The dataset also does not contain journey purpose, traffic conditions or road routes. The suggested explanations for commuting, nightlife and holiday activity are therefore interpretations of the observed patterns, not causal conclusions. Finally, each named holiday is represented by only one date in the six-month period, so its result should not be generalised to every year.

6 Conclusion

The cleaning process removed 1.04% of records and retained 1,443,535 plausible trips. Friday had the highest average daily activity. Weekday demand was shaped by commuting, while weekend demand shifted towards late-night travel. Memorial Day showed the largest reduction relative to its normal weekday, whereas Valentine's Day was busier than a typical Sunday. Trip speeds were highest in the early morning and lowest during the congested daytime period. Together, the results show that both trip volume and travel speed follow clear weekly and hourly patterns.

References

[1] Kaggle. New York City Taxi Trip Duration. https://www.kaggle.com/c/nyc-taxi-trip-duration.